

Multiple Choice (10 marks)

1. Which ratio CANNOT form a proportion with 8 over 18?
 - (a) 46 over 108
 - (b) 12 over 27
 - (c) 60 over 135
 - (d) 4 over 9
2. Three points are chosen randomly and independently on a circle. What is the probability that all three pairwise distances between the points are less than the radius of the circle?
 - (a) $\frac{1}{20}$
 - (b) $\frac{1}{32}$
 - (c) $\frac{1}{12}$
 - (d) $\frac{1}{16}$
3. We roll a fair 6-sided die 5 times. What is the probability that we get a 6 in at most 2 of the rolls?
 - (a) $\frac{125}{648}$
 - (b) $\frac{25}{648}$
 - (c) $\frac{625}{648}$
 - (d) $\frac{1}{648}$
4. The first number in a number pattern is 28. The pattern rule is to add 14 to get the next number in the pattern. If the pattern continues, which statement is true?
 - (a) All the numbers in the pattern can be divided equally by 10.
 - (b) All the numbers in the pattern can be divided equally by 4.
 - (c) All the numbers in the pattern can be divided equally by 8.
 - (d) All the numbers in the pattern can be divided equally by 7.
5. What is the sum of all integer solutions to $|n| < |n-3| < 9$?
 - (a) -9
 - (b) -14
 - (c) 14
 - (d) 9

True / False (10 marks)

Answer True or False

6. True or False: The number 47,520 rounds to 47,000 when rounded to the nearest thousand.
7. True or False: A worker producing parts at a constant rate can make 110 parts in 35 hours if she
8. True or False: The maximum value of the function $f(x) = xe^{-x}$ is e .
9. True or False: The volume of the solid with a square cross-section and a base bounded by $x^2 =$
10. True or False: The number of heads in three tosses of a coin is not a suitable scenario for a binomial

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. In how many ways can 9 people sit around a round table? (Two seatings are considered the same if one is a rotation of the other.)
12. If there is a 70% chance of rain on a given day, explain how you would determine the probability of it not raining, and analyze the implications of weather probabilities in daily decision-

End of Paper